IV B.Tech I Semester

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23A30702b	**HUMAN-COMPUTER INTERACTION (Professional Elective-IV)**	**3**	**0**	**0**	**3**

Course Objectives:

- To understand the fundamentals of Human-Computer Interaction.
- To design user-centric interfaces and systems with usability and accessibility in mind.
- To explore cognitive models and interaction design processes.
- To develop knowledge in evaluation techniques and prototyping.
- To introduce the latest advancements in HCI including AR/VR, gesture, and voice-based interfaces.

Course Outcomes:

After completing the course, students will be able to:

- Understand the principles and concepts of Human-Computer Interaction.
- Apply design techniques to create usable, intuitive interfaces.
- Analyze and evaluate user interfaces through cognitive and empirical methods.
- Implement effective prototypes and interaction models.
- Explore emerging trends in multimodal and intelligent interfaces.

UNIT I: Introduction to HCI & Human Factors

Introduction to HCI: History, Importance, and Applications – (U), Human Factors: Perception, Cognition, and Memory – (U), Models of Human Behavior: Norman's Model, Fitts' Law – (U), Ergonomics and Accessibility in Design – (U), Visual, Auditory, and Haptic Modalities – (U), Interaction Styles: Command-line, WIMP, Direct Manipulation – (R), Principles of Good Design: Consistency, Feedback, Affordance – (U), Case Study: Bad vs Good Interface Design – (U)

UNIT II: Design Process and User-Centered Design

Design Thinking and User-Centered Design – (U), Requirement Gathering: Task Analysis, Personas, Scenarios – (A), Information Architecture: Navigation and Flow – (A), Interface Design Guidelines (Shneiderman, Nielsen) – (U), Storyboarding and Wireframing – (A), Low-Fidelity vs High-Fidelity Prototyping – (A), Interaction Design Patterns – (A), Case Study: Designing an ATM Interface – (AN)

UNIT III: Evaluation Techniques and Usability Testing

Types of Evaluation: Formative vs Summative – (U), Usability Metrics: Learnability, Efficiency, Memorability – (A), Cognitive Walkthrough and Heuristic Evaluation – (A), A/B Testing and Controlled Experiments – (A), Eye Tracking and Think-Aloud Protocols – (A), User Feedback: Surveys and Interviews – (A), Quantitative and Qualitative Data Analysis – (AN), Case Study: Usability Analysis of a Mobile App – (E)

UNIT IV: Interface Technologies and Prototyping Tools

Graphical User Interfaces (GUI) and Web Interfaces – (U), Mobile and Touch Interfaces – (U), Voice-Based Interfaces (VUI) and Chatbots – (A), Gesture-Based Interaction and Wearables – (U), Augmented and Virtual Reality Interfaces – (A), Rapid Prototyping Tools: Figma, Adobe XD, Axure – (A), HTML/CSS/JS for UI Prototyping – (A), Case Study: Building a Prototype for a Fitness App – (A)

UNIT V: Advanced Topics in HCI and Future Trends

Brain-Computer Interfaces (BCI) – (U), Multimodal Interaction: Combining Voice, Gesture, Gaze – (U), Intelligent Interfaces using AI – (U), Emotion Recognition and Affective Computing – (U), Ethics in HCI: Privacy, Bias, and Design for Inclusion – (E), Accessibility Standards: WCAG, ADA – (A), Gamification and Persuasive Design – (AN), Capstone Project: Designing an Inclusive Learning App – (C)

Textbooks:

1. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, "Human-Computer Interaction", Pearson Education, 4th Edition.
2. Ben Shneiderman et al., "Designing the User Interface: Strategies for Effective Human-Computer Interaction", Pearson.
3. Donald A. Norman, "The Design of Everyday Things", Basic Books.

Reference Books:

1. Jenny Preece, Yvonne Rogers, Helen Sharp, "Interaction Design: Beyond Human-Computer Interaction", Wiley.
2. Jeff Johnson, "Designing with the Mind in Mind", Morgan Kaufmann.
3. Steven Krug, "Don't Make Me Think", New Riders.

Online Learning Resources:

- Coursera: Human-Centered Design by UC San Diego (https://www.coursera.org/learn/human-computer-interaction)
- edX: HCI by Georgia Tech (https://www.edx.org/course/human-computer-interaction)
- MIT OpenCourseWare: Human-Computer Interaction (https://ocw.mit.edu/)
- Interaction Design Foundation: https://www.interaction-design.org/
- Figma Design Tutorials: https://www.figma.com/resources/learn-design/